

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	20 February 2026
Team ID	LTVIP2026TMIDS34731
Project Name	HouseHunt-Finding Your Prefect Rental Home-
Maximum Marks	4 Marks

Technical Architecture:

1. Overview

HouseHunt is a MERN-stack based rental property web application designed to simplify the process of finding and renting homes. The platform connects **Renters, Property Owners, and Admins** in a secure and structured environment.

2. Components and Technologies Used

Component	Technology Used	Description
Frontend	React.js, Tailwind CSS	User interface and page routing
Backend	Node.js, Express.js	Server-side logic and API handling
Database	MongoDB (Mongoose)	Storing users, books, and orders
Authentication	JWT (JSON Web Token)	Secure user login and route protection
Version Control	Git & GitHub	Codebase management and collaboration
Package Management	npm (Node Package Manager)	Dependency management

Charting Library	Recharts.js	Charts for dashboards (admin/seller)
Hosting (Optional)	Render / Vercel / MongoDB Atlas	Hosting frontend/backend and database

3. Application Characteristics

Characteristic	Details
Architecture	MERN Stack (MongoDB, Express.js, React.js, Node.js)
Design Pattern	Modular MVC pattern (Separation of concerns)
API Communication	Axios with RESTful endpoints
Security	JWT Authentication, input validation
Scalability	Modular folder structure and route-based role access
UI Framework	Tailwind CSS for consistent responsive design